COMPUTATIONAL ECONOMICS
AS THE MENGERIAN COMPOSITIVE METHOD

"TO THE NAIVE MIND THAT CAN CONCEIVE OF ORDER ONLY AS THE PRODUCT OF DELIBERATE ARRANGEMENT, IT MAY SEEM ABSURD THAT ***IN COMPLEX CONDITIONS*** *ORDER, AND* ***ADAPTATION TO THE UNKNOWN, CAN BE ACHIEVED*** *MORE EFFECTIVELY* ***BY DECENTRALIZING DECISIONS****."*

FRIEDRICH HAYEK [1988]: *THE FATAL CONCEIT.*

JIRI NOHEJL (GEORGE MASON UNIVERSITY)

COMPUTATIONAL ECONOMICS COULD BE ARGUED AS A PATHWAY TOWARDS POST-WALRASIAN FRAMEWORK (COLANDER, 2006), BUT HERE IT IS DESCRIBED AS THE ONE WHICH IS **EMBEDDED IN A "NEO-MENGERIAN" COMPOSITIVE METHOD.**

COMPOSITIVE METHOD OF MENGER AND HAYEK

Carl Menger (1871) was a leading figure in Marginalist Revolution proposing method which differentiated his approach from his contemporaries.

> "[Menger's] economics in its substantive content was **disequilibrium economics**, it was also in a broad sense institutional economics. […] For the performance of this task what is required is not the mathematical method, but a **method of process analysis tracing the complex phenomena of the social economy to the underlying forces at work**."
>
> William Jaffé [1972]: *Menger, Jevons and Walras De-Homogenized.*

Friedrich Hayek (1955) embraced Menger's method and emphasized its "**compositive**" character drawing on an anecdote in which Menger himself so adjusted "deductive" used in Schmoller's review of his *Investigations into the Method of the Social Sciences.*

COMPUTING IN/OUT ECONOMIC SCIENCE

> "The understanding [Miller and Drexler] exhibited of Hayek seemed to me to be far more sophisticated than that of most economists, and the number of ways in which **contemporary computer science** appeared to be **closely related to Hayekian themes** was astonishing."
>
> Don Lavoie et al. [1990]: *High-Tech Hayekians: Some Possible Research Topics in the Economics of Computation.*

Agoric Papers (Miller and Drexler, 1988) are exploring a similarities of Hayekian "knowledge problem" and problems of distributed computing in relation to "market" and "centrally-planned" paradigms.

Research questions if distributed (parallel) computation should be like economic system (Huberman & Hogg, 1995) or if economics is like distributed computation (Axtell, 2002) leads to questions of correspondence and complementarity of these research programs.

CONNECTIONISM, COMPOSITION AND NETWORKS

> "[T]he neo-Walrasian program conceptualizes society as a **field** In contrast, the neo-Mengerian program conceptualizes it as a **network**; the properties of a network, in contrast to those of a field, vary with its constitutive **structure**.
>
> Richard E. Wagner [2010]: *Change within Permanence: Time and the Bivalent Logic of Economic Analysis.*

Fully **connected graph** of Walrasian framework is contrasting with a Mengerian **compositive structures of evolving networks** in which position of links and their weight matter.

"NEURONS WHICH FIRE TOGETHER – WIRE TOGETHER."

The idea of **compositive market structure** is here correspondent with the **connectionist approach to structure of mind** (Hebb, 1949; Hayek, 1952).

"THE **PHENOMENA OF THE MARKET** ARE AT THE SAME TIME **PHENOMENA OF THE MIND,** AND THEY MUST BE EXPLAINED ACCORDINGLY."

SMALL, ALBION. W. (1924). *ORIGINS OF SOCIOLOGY.*

ARTIFICIAL INTELLIGENCE: MODERN SCALE

LLMs can be understood as artificial neural networks with massive amounts of computational power utilized in training driven by the mantra “scale is what you need”.

> “[T]he current debate suggests a fascinating divergence in how to think about understanding in intelligent systems, in particular the contrast between mental models that rely on **statistical correlations** and those that rely on **causal mechanisms**.”
>
> Mitchell, Melanie & Krakauer, David C. [2023]: *The Debate over Understanding in AI’s Large Language Models.*

(1) These challenges somehow resembles Menger-Schmoller dispute over method.

(2) Alternatively, we can treat LLMs as depositors of knowledge capital which were created by massively applying computational labor – model pre-training

(3) We can analogize from ANN models to enhance compositive computational economics.

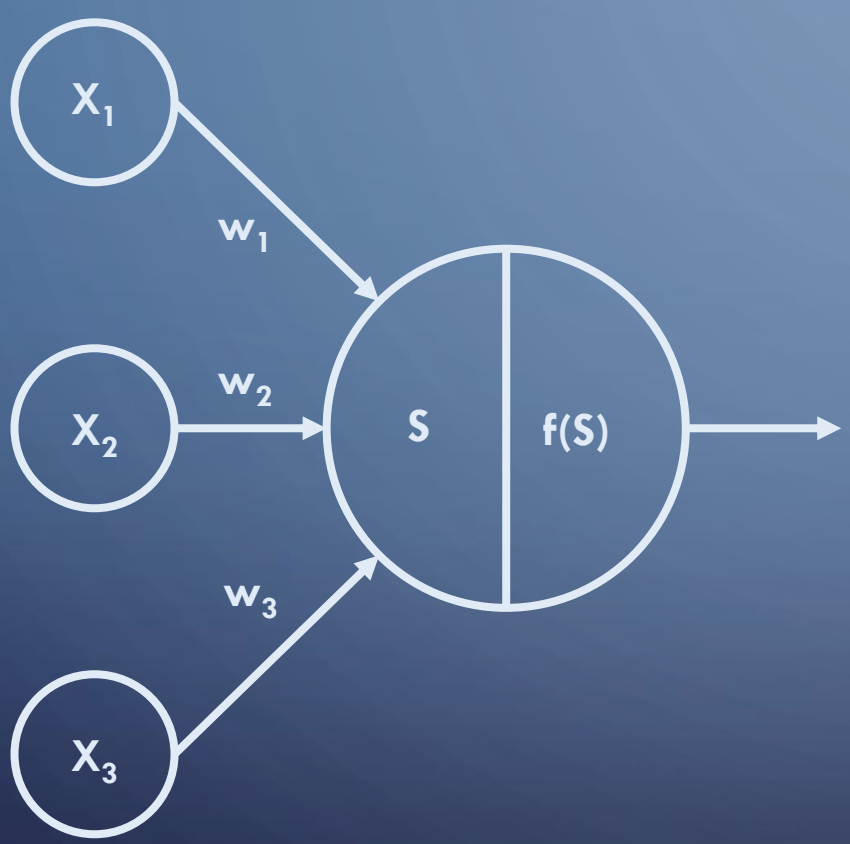
FROM LLM BACK TO NETWORKS OF PERCEPTRONS

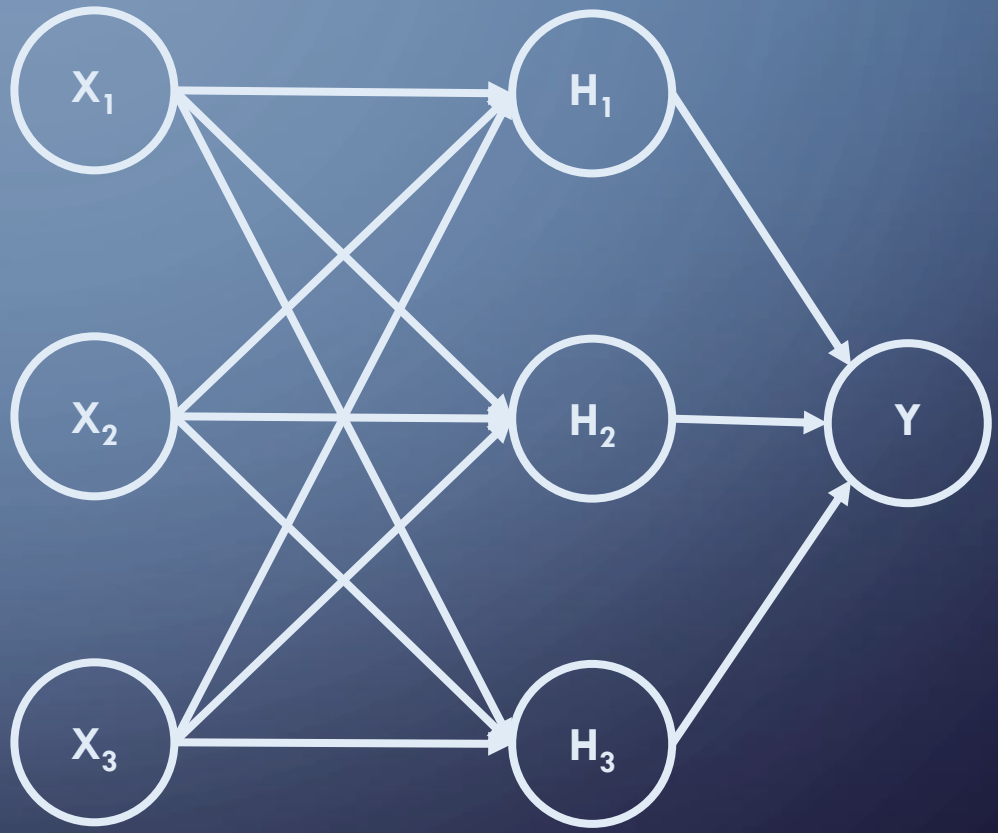
The idea of Perceptron is that complex structures of interconnected connected elements endowed with rather limited computation capabilities lead to emergent learning.

> "The contributions of this group should perhaps be considered as suggestions of what to look for and investigate, rather than as finished theoretical systems in their own right. Seen from this viewpoint, **the most suggestive work, from the standpoint of the following theory, is that of Hebb and Hayek**."
>
> Frank Rosenblatt [1958]: *The Perceptron: A Probabilistic Model For Information Storage and Organization in the Brain.*

Artificial intelligence explore potentials of parallel and layered computing by which it creates models useful to study parallel and layered computing of the market.

FROM LLM BACK TO NETWORKS OF PERCEPTRONS





ARTIFICIAL INTELLIGENCE: SOLUTION OR MODEL

Computational economics can be understood as approach drawing from **economics** and **computer shift** is computation (1) a tool or (2) a method.

(1) DSGE model within Heterogenous Agents New Keynesian (HANK) framework can benefit from Monte Carlo style computational simulation or from a specific DSGE models pre-trained AI model.

(2) Methods used to computationally simulate intelligence are form of "**generative social science**" creating worlds of ***in silico*** interactions (Epstein, 2012).

First approach does not shift methods away from Walrasian paradigm but the other embrace economics as a composition of interacting elements.

Hayek insight in *Economics and Knowledge* (1937) of moving away from general equilibrium rezoning towards an ecology of plans can be realized by composition of (AI) agents with heterogeneity and intentionality.

"[T]HE JOB OF **WALRASIAN AUCTIONEER** BECOMES EXPONENTIALLY HARDER AS THE NUMBER OF COMMODITIES INCREASES. THIS **IS COMPLETELY UNREALISTIC** AND MAKES THE **MODEL [OF GENERAL EQUILIBRIUM]** APPEAR TO BE **COMPUTATIONALLY NAIVE.**"

ROBERT AXTELL [2016]: *HAYEK ENRICHED BY COMPLEXITY ENRICHED BY HAYEK.*

COMPOSITIVE METHOD: MIND AND MONEY

> "Interesting things in markets happen because of their temporal phenomena, they happen within **departures from equilibrium** […] that, after all, is **where the money is made**."
>
> W. Brian Arthur [2013]: *Complexity Economics: A Different Framework for Economic Though.*

Most explicit application of Menger's application of compositive method is his theoretical treatment of emergence of money in *On the Origin of Money* (1892).

Imposing correspondence to Hayek's connectionism of the structure of mind:

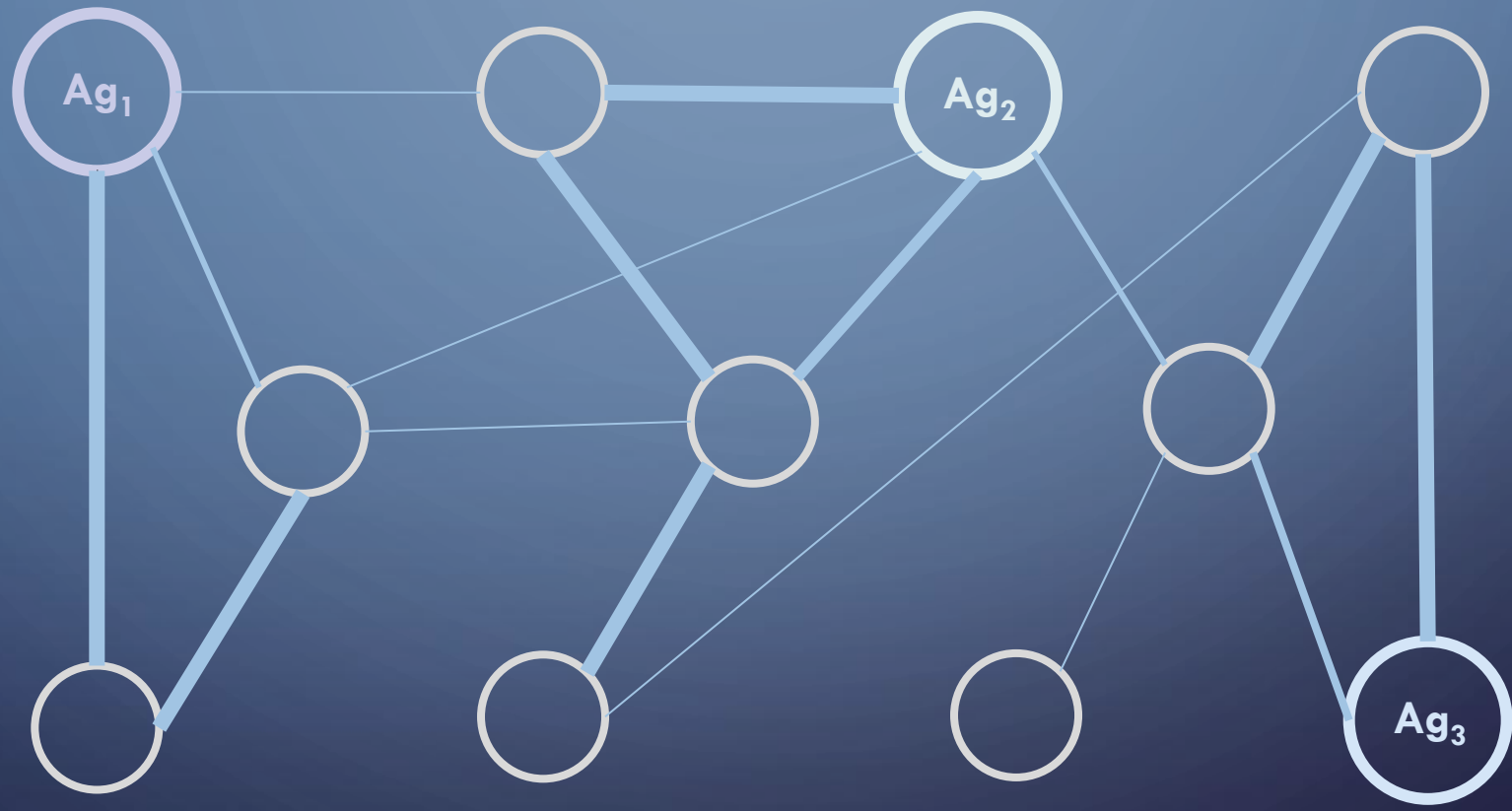
"Agents which Exchange Together – Connect Together."

In *The Sensory Order* (1952) Hayek differentiates: (1) "physical order", (2) "neural order" and (3) "phenomenal order" which analogically leads to differentiation of **real** (material) **and nominal** (informational) **order of economic system**.

COMPUTATIONAL MONETARY SYSTEMS

$[0,8*1/P_A; 0,2*1/P_B; 0,0*1/P_C]$

$[0,1*1/P_A; 0,7*1/P_B; 0,2*1/P_C]$



$[0,0*1/P_A; 0,0*1/P_B; 1,0*1/P_C]$

COMPUTATIONAL MONETARY SYSTEMS

> "[A]different way of thinking about the economy. It sees the economy **not as a system in equilibrium** but as one in motion, **perpetually 'computing' itself** - perpetually constructing itself anew."
>
> W. Brian Arthur [2013]: *Complexity Economics: A Different Framework for Economic Thought.*

In contrast to DSGE modeling where in equilibrium money "does not matter" in suggested framework money and monetary prices are establishing network topology.

Analogy of market with computational intelligence allows modeling of money as decentralize information systems adapting to ever changing conditions.

Applications: (1) injection effects analysis, (2) topology of network effects and switching costs, (3) Bitcoin on-chain / off-chain dynamics.

"MENGER WROTE LONG BEFORE SUCH TECHNIQUES AS AGENT-BASED COMPUTATIONAL MODELING HAD ARISEN, BUT IT'S CLEAR THAT **MENGER'S THEORETICAL INTUITIONS WOULD HAVE SUPPORTED EMERGENT AND GENERATIVE STYLES OF THEORIZING.**"

RICHARD E. WAGNER [2010]: *CHANGE WITHIN PERMANENCE*.

THANK YOU FOR YOUR ATTENTION!

Jiri Nohejl
Department of Economics
George Mason University
jnohejl@gmu.edu